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Amir Reza Vazifeh

EDUCATION

Princeton University
*Ph.D., **Electrical & Computer Engineering*** *2023 – 2028*
- GPA 3.9/4.0; Ph.D. Fellowship in Natural Sciences and Engineering
- **Research**: Computational Imaging & Optics, Computer Vision, AI in Healthcare, Signal Processing

Sharif University of Technology
*B.S., **Electrical Engineering**, Minor in **Physics*** *2018 – 2022*
- GPA 18.83/20; Top 10% of class; Academic Achievement Award (2019–2020); 12^{th}/134K in National Entrance Exam
- **Research**: Biomedical Signal Processing, AI in Healthcare, Wearable Sensors, Circuit Design, Embedded Systems

EXPERIENCE

Princeton University (Biomedical Imaging Group) Princeton, NJ
Graduate Researcher *(Advisor: Prof. Jason Fleischer; Collaborator: Prof. Felix Heide)* *2024 – Present*
- Developing unsupervised video-based image reconstruction in fiber endoscopy using implicit neural representations
- Creating a framework for manifold learning and alignment for personalized multi-lead ECG analysis
- Implementing an algorithm for snapshot hyperspectral imaging with implicit neural representations

Waters Corporation (Industry PhD Partnership) Remote, US
Computer Vision and Machine Learning Researcher *2024 – Present*
- Designing and implementing a deep learning cell counting and viability detection system on brightfield and phase microscopy, robust to focal plane variations and staining strategies
- Driving Waters' next-generation imaging platform; recognized by senior leadership up to SVP level

Institute of Science and Technology Austria Vienna, Austria
Research Intern *(among top 3% of applicants) (Advisor: Prof. Sandra Siegert)* *Summer 2023*
- Implemented 3D instance segmentation of microglia cells using classical computer vision techniques

EPFL (Integrated Neurotechnologies Lab) Geneva, Switzerland
Research Intern *(among top 2% of applicants) (Advisor: Prof. Mahsa Shoaran)* *Summer 2022*
- Analyzed phase-amplitude cross-frequency coupling in EEG signals

Sharif University of Technology (Biosen Group) Tehran, Iran
Undergraduate Researcher *(Advisors: Profs. Mohammad Fakharzadeh & Daryoosh Vashayee)* *2020 – 2022*
- Designed a low-power wearable ECG patch for long-term cardiac monitoring with ML-based arrhythmia detection
- Researched energy harvesting materials and techniques for wearable medical devices

PATENTS

- **Cell Viability Assessment** — U.S. Non-Provisional Patent Application
  Assignee: Waters Corporation — *Filed:* June 2026
  Inventors: Christian Zeigler, Amir Reza Vazifeh, Sornanathan Meyyappan, Richard Jeske, Jason W. Fleischer

PUBLICATIONS

- **Under Review**: AI-Enabled Label-Free Cell Viability Assessment from Brightfield Imaging, A. R. Vazifeh et al. *Frontiers in Optics*, 2026
- Manifold Alignment for Label-Free Cell Phenotyping in Multimodal Microscopy, A. R. Vazifeh, J. W. Fleischer. *Optica Biophotonics Congress — Microscopy, Histopathology and Analytics*, 2026
- Seeing Through Fibers: Unsupervised Image Reconstruction in Fiber Bundle Imaging Systems, A. R. Vazifeh et al. *Optics Express*, 2026
- Manifold Learning for Personalized and Label-Free Detection of Cardiac Arrhythmias, A. R. Vazifeh, J. W. Fleischer. *Informatics in Medicine Unlocked*, 2026
- An Open-Source Retrospective Analysis of Hypertrophic and Dilated Cardiomyopathy Using Machine Learning and Electrocardiogram Data, A. Altintepe et al. *Diagnostics*, 2026
- Snapshot Hyperspectral Imaging via Compressive Sensing and Implicit Neural Representation, J. Boondicharern, A. R. Vazifeh, J. W. Fleischer. *Optica Imaging Congress — Computational Optical Sensing and Imaging*, 2025
- Advancing Personalized Healthcare: Progress in Energy Harvesting Materials of Self-Powered Wearable Devices, P. Bhatnagar et al. *Progress in Materials Science* (Impact Factor: 40), 2023
- Design and Implementation of an Ultralow-Power ECG Patch and Smart Cloud-Based Platform, B. Baraeinejad et al. *IEEE Transactions on Instrumentation and Measurement*, 2022

TEACHING & MENTORSHIP

Princeton University Princeton, NJ
Teaching Assistant & Research Mentor *2024 – 2026*

- TA for Biomedical Imaging (ECE 452) and Machine Learning & Pattern Recognition (ECE 435/535); research mentor

Sharif University of Technology Tehran, Iran
Teaching Assistant *2020 – 2022*

- TA for Introduction to Machine Learning, Signals & Systems, Computer Architecture, and Applied Electronics in Bioengineering

SKILLS

Programming & ML — Python (PyTorch, TensorFlow, Sklearn, OpenCV, NumPy, SciPy), C++
Cloud & Data — AWS (S3, EC2), large-scale data analysis, distributed/parallel computing
Hardware — Circuit analysis & board design (Altium, HSPICE, Proteus), HDL (MIPS Assembly, Verilog)
Other — Git, LaTeX, HTML, CSS; reviewer for *IEEE Journal of Biomedical and Health Informatics* (JBHI)